# APEX NCAA FOOTBALL - DAILY GRADER

## TARGET SLATE 2026-10-20

Layer 1 (sections A-F): yesterday, season record, probability quality against FanDuel, calibration, ratings, integrity. Layer 2 (appendices 1-5): technical evidence, hashes and diagnostics.

| FIELD | PHYSICAL VALUE |
|---|---|
| TARGET SLATE | 2026-10-20 |
| GENERATED ET | 2026-10-21T07:56:53.710494-04:00 |
| ATS MODEL | D2 / 09c33b70af53b0a53e2127b16b00d23d37b8de692edc76f1ab83b1a25109d2e0 |
| TOTALS MODEL | TOTALS-V5-RECONCILED-POOL / e66c7f0d35a0827a93458b84e083d3d0fa4e54724af82470a674acb961c2243f |
| CANONICAL ISSUANCE | 0d15a3c6f93c346e5654127233681033508f8c3a2362d4e2576a3a16b873348c |
| GRADING COMPLETION RECEIPT | 0b8e68f93b6f98d82de513d7d3b1b70dc95766173f8c9bac66b075af663511ad |
| REPORT SCHEMA | apex.ncaaf.daily_grader_evidence_book.v5.two_layer |

### POPULATIONS USED IN THIS REPORT

| POPULATION | DEFINITION | SLATES | GAMES | ATS | TOTALS | POSITIONS |
|---|---|---|---|---|---|---|
| TARGET SLATE | all physical positions issued for 2026-10-20 | 1 | 3 | 3 | 3 | 6 |
| of which EXCLUDED POST-KICKOFF | issued after kickoff; visible, never scored | — | — | — | — | 0 |
| of which PROSPECTIVE ELIGIBLE | enter records and proper scores | — | — | 3 | 3 | 6 |
| PHYSICAL POSITION-EVIDENCE CORPUS | every immutable issued position through 2026-10-20 | 29 | 509 | 509 | 509 | 1018 |
| minus EXCLUDED POST-KICKOFF | issued after kickoff; visible, never scored | 2 | 16 | 16 | 16 | 32 |
| = RECORD ELIGIBLE GRADED CORPUS | the only population in records, calibration and scores | 29 | 493 | 493 | 493 | 986 |
| settled (W+L+P+V) | — | — | — | 493 | 493 | 986 |
| pending | official result not final | — | — | 0 | 0 | 0 |

RECONCILIATION · physical 1018 – excluded post-kickoff 32 = record eligible 986 (ATS 509 – 16 = 493; TOTALS 509 – 16 = 493). Excluded by slate: 2026-09-25 (2), 2026-09-26 (30).

### HOW TO READ

Records use W, L, P (push), V (void) and pending. Win rate is W/(W+L).

Every grade settles the exact immutable issued proposition. No engine is rerun and no later FanDuel line replaces the issued line.

Ratings (WEAK / MODERATE / STRONG / ELITE) are labels on the selected win probability, assigned at issuance by the rating definition sealed with each position (tier_policy_sha256) and retained exactly as issued, except a separately recorded and verified pre-kickoff public revision. Original ratings and revision evidence remain in the position export; no grader run computes new ratings. The definition changed during the season, so a label has one meaning only within one definition and engine. Target-slate positions were rated under SLATE-QUARTILES-V3. A rating is relative confidence within its definition and engine, not a win probability; the model probability is shown separately (APEX P). Ratings are evaluated, not assumed, in section E, grouped by definition.

RATING DEFINITION FOUR-TIER-V1 · APEX_NCAAF_CANONICAL_FOUR_TIER_V1 · sha256 65fbd6df6e5528e7… · issued 2026-08-29 … 2026-09-03 · 38 positions · MODERATE / STRONG / ELITE lower cut points on the selected win probability: ATS 0.5300 / 0.5600 / 0.6000 · TOTALS 0.5300 / 0.5600 / 0.6000 (same cut points for both engines).

RATING DEFINITION SHARED-51.75 · APEX_MAX_CALIBRATED_WIN_PROBABILITY_RATING_CONTRACT_V3_PRESENTATION_MIRROR · sha256 c7d506cdb615c0db… · issued 2026-09-04 … 2026-09-19 · 476 positions · MODERATE / STRONG / ELITE lower cut points on the selected win probability: ATS 0.5175 / 0.5375 / 0.5575 · TOTALS 0.5175 / 0.5375 / 0.5575 (same cut points for both engines).

RATING DEFINITION SHARED-52/54/58 · APEX_MAX_CALIBRATED_WIN_PROBABILITY_RATING_CONTRACT_V3_PRESENTATION_MIRROR · sha256 a69977c71f3df62b… · issued 2026-09-24 … 2026-10-09 · 282 positions · MODERATE / STRONG / ELITE lower cut points on the selected win probability: ATS 0.5200 / 0.5400 / 0.5800 · TOTALS 0.5200 / 0.5400 / 0.5800 (same cut points for both engines).

RATING DEFINITION ENGINE-V2 · APEX_NCAAF_ENGINE_SPECIFIC_RATING_BANDS_V2_20261009 · sha256 a7acf120b9b1c889… · issued 2026-10-10 … 2026-10-16 · 112 positions · MODERATE / STRONG / ELITE lower cut points on the selected win probability: ATS 0.5120 / 0.5318 / 0.5460 · TOTALS 0.5044 / 0.5111 / 0.5209 (engine-specific cut points at the 40th / 78th / 95th percentiles of a 100-position reference per engine (Oct 3 and Oct 10 rescore)).

RATING DEFINITION SLATE-QUARTILES-V3 · APEX_NCAAF_SLATE_RELATIVE_QUARTILES_V3 · sha256 6de90770392f7efd… · issued 2026-10-17 … 2026-10-20 · 110 positions · MODERATE / STRONG / ELITE lower cut points on the selected win probability: ATS slate-specific (retained per position) · TOTALS slate-specific (retained per position) (strict-lower percentile rank within each engine's scoreable slate; rank boundaries 0.25 / 0.50 / 0.75; probability cut points vary by slate and are retained per position; ties share a tier).

NCAA full-game spread and total settlement includes overtime unless the applicable issued-market rules state otherwise. A whole-number line landing on the result is a push.

KNOWN EVIDENCE LIMITATION · FANDUEL_SETTLEMENT_RULE_SOURCE=UNRECORDED_JURISDICTION_DETAIL: the jurisdiction-specific FanDuel settlement rule text in force for each issued market is not recorded in the evidence; every grade settles the issued line and price against the official final score.

# A. YESTERDAY — TARGET-SLATE POSITION LEDGER (2026-10-20)

6 physical positions on 3 games; 0 excluded post-kickoff. FD P = same-side FanDuel no-vig probability from the issuance-time quote (two-way multiplicative normalization); Δ = APEX − FD.

| G | MATCHUP | KICK ET | MKT | SELECTION | LINE | PRICE | APEX P | FD P | Δ | RATING | FINAL | RES |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Middle Tennessee at Florida International | Tue 19:00 | ATS | Middle Tennessee | +10.5 | -120 | 51.5% | 51.9% | -0.4 | MODERATE | 26-23 (A-H) | W |
| 1 | Middle Tennessee at Florida International | Tue 19:00 | TOTALS | UNDER | 45.5 | -110 | 50.1% | 50.0% | +0.1 | WEAK | 26-23 (A-H) | L |
| 2 | Missouri State at Delaware | Tue 19:00 | ATS | Delaware | -2.5 | -118 | 52.0% | 51.5% | +0.5 | STRONG | 38-61 (A-H) | W |
| 2 | Missouri State at Delaware | Tue 19:00 | TOTALS | OVER | 53.5 | -110 | 50.3% | 50.0% | +0.3 | MODERATE | 38-61 (A-H) | W |
| 3 | South Alabama at Marshall | Tue 19:30 | ATS | Marshall | +2.5 | -115 | 51.2% | 51.1% | +0.2 | WEAK | 13-20 (A-H) | W |
| 3 | South Alabama at Marshall | Tue 19:30 | TOTALS | UNDER | 65.5 | -110 | 50.9% | 50.0% | +0.9 | STRONG | 13-20 (A-H) | W |

TARGET-SLATE RECORD · ATS 3W-0L-0P-0V (100.0%) · TOTALS 2W-1L-0P-0V (66.7%) · COMBINED 5W-1L-0P-0V (83.3%) · pending 0 · excluded post-kickoff 0. All positions derive from immutable issuance 0d15a3c6f93c346e…; no closing-line substitution.

# B. SEASON — RECORD ELIGIBLE GRADED CORPUS

29 physical slates 2026-08-29 … 2026-10-20; 986 record eligible positions. Win rate is an operational outcome, not the forecast evaluation (see C and D).

| ENGINE | N | W | L | P | V | PENDING | SETTLED | WIN % |
|---|---|---|---|---|---|---|---|---|
| ATS | 493 | 259 | 234 | 0 | 0 | 0 | 493 | 52.5% |
| TOTALS | 493 | 253 | 240 | 0 | 0 | 0 | 493 | 51.3% |
| COMBINED | 986 | 512 | 474 | 0 | 0 | 0 | 986 | 51.9% |
| COMBINED · FOUR-TIER-V1 · WEAK | 23 | 16 | 7 | 0 | 0 | 0 | 23 | 69.6% |
| COMBINED · FOUR-TIER-V1 · MODERATE | 5 | 1 | 4 | 0 | 0 | 0 | 5 | 20.0% |
| COMBINED · FOUR-TIER-V1 · STRONG | 3 | 1 | 2 | 0 | 0 | 0 | 3 | 33.3% |
| COMBINED · FOUR-TIER-V1 · ELITE | 7 | 6 | 1 | 0 | 0 | 0 | 7 | 85.7% |
| COMBINED · SHARED-51.75 · WEAK | 195 | 108 | 87 | 0 | 0 | 0 | 195 | 55.4% |
| COMBINED · SHARED-51.75 · MODERATE | 190 | 95 | 95 | 0 | 0 | 0 | 190 | 50.0% |
| COMBINED · SHARED-51.75 · STRONG | 71 | 37 | 34 | 0 | 0 | 0 | 71 | 52.1% |
| COMBINED · SHARED-51.75 · ELITE | 20 | 14 | 6 | 0 | 0 | 0 | 20 | 70.0% |
| COMBINED · SHARED-52/54/58 · WEAK | 125 | 65 | 60 | 0 | 0 | 0 | 125 | 52.0% |
| COMBINED · SHARED-52/54/58 · MODERATE | 85 | 33 | 52 | 0 | 0 | 0 | 85 | 38.8% |
| COMBINED · SHARED-52/54/58 · STRONG | 40 | 22 | 18 | 0 | 0 | 0 | 40 | 55.0% |
| COMBINED · SHARED-52/54/58 · ELITE | 0 | 0 | 0 | 0 | 0 | 0 | 0 | N/A |
| COMBINED · ENGINE-V2 · WEAK | 53 | 26 | 27 | 0 | 0 | 0 | 53 | 49.1% |
| COMBINED · ENGINE-V2 · MODERATE | 43 | 20 | 23 | 0 | 0 | 0 | 43 | 46.5% |
| COMBINED · ENGINE-V2 · STRONG | 15 | 8 | 7 | 0 | 0 | 0 | 15 | 53.3% |
| COMBINED · ENGINE-V2 · ELITE | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 100.0% |
| COMBINED · SLATE-QUARTILES-V3 · WEAK | 28 | 13 | 15 | 0 | 0 | 0 | 28 | 46.4% |
| COMBINED · SLATE-QUARTILES-V3 · MODERATE | 28 | 18 | 10 | 0 | 0 | 0 | 28 | 64.3% |
| COMBINED · SLATE-QUARTILES-V3 · STRONG | 28 | 14 | 14 | 0 | 0 | 0 | 28 | 50.0% |
| COMBINED · SLATE-QUARTILES-V3 · ELITE | 26 | 14 | 12 | 0 | 0 | 0 | 26 | 53.8% |

| SLATE | ATS N | ATS W-L-P | ATS WIN % | TOT N | TOT W-L-P | TOT WIN % | EXCL |
|---|---|---|---|---|---|---|---|
| 2026-08-29 | 8 | 5-3-0 | 62.5% | 8 | 6-2-0 | 75.0% | 0 |
| 2026-09-03 | 11 | 9-2-0 | 81.8% | 11 | 4-7-0 | 36.4% | 0 |
| 2026-09-04 | 7 | 4-3-0 | 57.1% | 7 | 4-3-0 | 57.1% | 0 |
| 2026-09-05 | 68 | 39-29-0 | 57.4% | 68 | 33-35-0 | 48.5% | 0 |
| 2026-09-06 | 3 | 0-3-0 | 0.0% | 3 | 2-1-0 | 66.7% | 0 |
| 2026-09-07 | 1 | 1-0-0 | 100.0% | 1 | 1-0-0 | 100.0% | 0 |
| 2026-09-10 | 1 | 1-0-0 | 100.0% | 1 | 0-1-0 | 0.0% | 0 |
| 2026-09-11 | 5 | 3-2-0 | 60.0% | 5 | 2-3-0 | 40.0% | 0 |
| 2026-09-12 | 80 | 43-37-0 | 53.8% | 80 | 44-36-0 | 55.0% | 0 |
| 2026-09-17 | 1 | 1-0-0 | 100.0% | 1 | 1-0-0 | 100.0% | 0 |
| 2026-09-18 | 3 | 1-2-0 | 33.3% | 3 | 1-2-0 | 33.3% | 0 |
| 2026-09-19 | 69 | 33-36-0 | 47.8% | 69 | 40-29-0 | 58.0% | 0 |
| 2026-09-24 | 1 | 0-1-0 | 0.0% | 1 | 1-0-0 | 100.0% | 0 |
| 2026-09-25 | 4 | 2-2-0 | 50.0% | 4 | 2-2-0 | 50.0% | 2 |

| SLATE | ATS N | ATS W-L-P | ATS WIN % | TOT N | TOT W-L-P | TOT WIN % | EXCL |
|---|---|---|---|---|---|---|---|
| 2026-09-26 | 49 | 22-27-0 | 44.9% | 49 | 25-24-0 | 51.0% | 30 |
| 2026-10-01 | 2 | 1-1-0 | 50.0% | 2 | 1-1-0 | 50.0% | 0 |
| 2026-10-02 | 3 | 2-1-0 | 66.7% | 3 | 1-2-0 | 33.3% | 0 |
| 2026-10-03 | 54 | 29-25-0 | 53.7% | 54 | 22-32-0 | 40.7% | 0 |
| 2026-10-06 | 1 | 1-0-0 | 100.0% | 1 | 0-1-0 | 0.0% | 0 |
| 2026-10-07 | 2 | 1-1-0 | 50.0% | 2 | 2-0-0 | 100.0% | 0 |
| 2026-10-08 | 4 | 1-3-0 | 25.0% | 4 | 3-1-0 | 75.0% | 0 |
| 2026-10-09 | 5 | 3-2-0 | 60.0% | 5 | 1-4-0 | 20.0% | 0 |
| 2026-10-10 | 46 | 19-27-0 | 41.3% | 46 | 24-22-0 | 52.2% | 0 |
| 2026-10-13 | 2 | 2-0-0 | 100.0% | 2 | 2-0-0 | 100.0% | 0 |
| 2026-10-14 | 2 | 1-1-0 | 50.0% | 2 | 1-1-0 | 50.0% | 0 |
| 2026-10-15 | 3 | 2-1-0 | 66.7% | 3 | 2-1-0 | 66.7% | 0 |
| 2026-10-16 | 3 | 0-3-0 | 0.0% | 3 | 2-1-0 | 66.7% | 0 |
| 2026-10-17 | 52 | 30-22-0 | 57.7% | 52 | 24-28-0 | 46.2% | 0 |
| 2026-10-20 | 3 | 3-0-0 | 100.0% | 3 | 2-1-0 | 66.7% | 0 |

# C. PROBABILITY QUALITY AGAINST FANDUEL — PAIRED, IDENTICAL OBSERVATIONS

Each row pairs the APEX issued probability with the same-side FanDuel no-vig probability from the quote captured at issuance, on the same game, market and settled W/L outcome. Lower is better. Δ = APEX − FanDuel (negative favours APEX).

| ENGINE | PAIRED N | APEX BRIER | FD BRIER | Δ BRIER | APEX LOGLOSS | FD LOGLOSS | Δ LOGLOSS | APEX < FD (BRIER) |
|---|---|---|---|---|---|---|---|---|
| ATS | 485 | 0.249936 | 0.250638 | -0.000701 | 0.693018 | 0.694423 | -0.001405 | 251 of 485 |
| TOTALS | 485 | 0.251200 | 0.248974 | +0.002225 | 0.695547 | 0.691095 | +0.004452 | 238 of 485 |

Record eligible W/L rows without a FanDuel comparator in their immutable grade (not in the paired table, listed by slate and sealed engine): ATS 8 [2026-08-29 (8, GLOBAL_EMPIRICAL_V1)]; TOTALS 8 [2026-08-29 (8, RIDGE_GAUSSIAN_A100_V1)]. FanDuel no-vig method as registered in each grade object: TWO_WAY_MULTIPLICATIVE_NORMALIZATION.

### TOTALS — RETAINED FULL-DISTRIBUTION SCORES (APEX ONLY)

| SCORE | N | VALUE | CONVENTION |
|---|---|---|---|
| PMF LOGSCORE (total PMF) | 490 | 4.081674 | retained integer-support PMF, read from the immutable grade |
| RPS | 490 | 0.034236 | retained integer-support PMF, read from the immutable grade |
| ISSUED PUSH PROBABILITY (mean) | 493 | 0.000000 | explicit in each sealed distribution |

No FanDuel full-distribution comparator exists: a single total line and two prices do not determine a distribution over totals, so none is constructed. The paired FanDuel comparison above is on the binary selected-side probability only.

# D. CALIBRATION (RECORD ELIGIBLE W/L ROWS)

Calibration (is p the realized frequency?) and discrimination (does p separate outcomes?) are reported separately. Brier = miscalibration − discrimination + uncertainty (Murphy decomposition on the PAV fit).

| ENGINE | N | MEAN ISSUED P | EMPIRICAL FREQ | CAL GAP | BRIER | LOGLOSS | MISCAL (MCB) | DISCRIM (DSC) | UNCERT (UNC) | AUC |
|---|---|---|---|---|---|---|---|---|---|---|
| ATS | 493 | 0.5203 | 0.5254 | +0.0051 | 0.249006 | 0.690759 | 0.001665 | 0.003064 | 0.249357 | 0.4791 |
| TOTALS | 493 | 0.5240 | 0.5132 | -0.0108 | 0.250937 | 0.695260 | 0.002014 | 0.003452 | 0.249826 | 0.4674 |

| ATS DIAGNOSTIC BIN (issued p) | N | W | L | MEAN P | EMPIRICAL | GAP |
|---|---|---|---|---|---|---|
| [0.50, 0.51) | 156 | 86 | 70 | 0.5051 | 0.5513 | +0.0461 |
| [0.51, 0.52) | 133 | 72 | 61 | 0.5152 | 0.5414 | +0.0261 |
| [0.52, 0.54) | 171 | 79 | 92 | 0.5283 | 0.4620 | -0.0664 |
| [0.54, 0.58) | 31 | 20 | 11 | 0.5497 | 0.6452 | +0.0955 |
| [0.58, 1.00) | 2 | 2 | 0 | 0.8848 | 1.0000 | +0.1152 |
| **TOTALS DIAGNOSTIC BIN (issued p)** | **N** | **W** | **L** | **MEAN P** | **EMPIRICAL** | **GAP** |
| [0.50, 0.51) | 157 | 86 | 71 | 0.5043 | 0.5478 | +0.0434 |
| [0.51, 0.52) | 122 | 64 | 58 | 0.5140 | 0.5246 | +0.0106 |
| [0.52, 0.54) | 119 | 53 | 66 | 0.5291 | 0.4454 | -0.0837 |
| [0.54, 0.58) | 82 | 41 | 41 | 0.5513 | 0.5000 | -0.0513 |
| [0.58, 1.00) | 13 | 9 | 4 | 0.6373 | 0.6923 | +0.0550 |

DIAGNOSTIC BINS ONLY · fixed reporting edges 0.50 / 0.51 / 0.52 / 0.54 / 0.58 on the issued probability, the same every day so bins compare across days. They are not rating cut points: each position's rating comes from its own sealed definition (HOW TO READ and section E). Small bins are shown with their N; nothing is pooled across engines.

# E. RATINGS — AS ISSUED, GROUPED BY RATING DEFINITION

Historical labels are immutable and are not recalculated. Verified pre-kickoff public revisions are identified in the position export with their originals retained. Each band is reported within one engine and one rating definition (the tier_policy_sha256 sealed at issuance); CUT is that band's lower cut point on the selected win probability. N is shown for every band; tiny bands are not averaged away.

| ENGINE | DEFINITION | BAND | CUT | N | W-L-P-V | WIN % | MEAN ISSUED P | EMPIRICAL | CAL GAP | BRIER | LOGLOSS | SCORE N |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ATS | FOUR-TIER-V1 | WEAK | — | 13 | 10-3-0-0 | 76.9% | 0.5080 | 0.7692 | +0.2613 | 0.24942 | 0.69199 | 13 |
| ATS | FOUR-TIER-V1 | MODERATE | 0.5300 | 2 | 1-1-0-0 | 50.0% | 0.5441 | 0.5000 | -0.0441 | 0.26227 | 0.71788 | 2 |
| ATS | FOUR-TIER-V1 | STRONG | 0.5600 | 2 | 1-1-0-0 | 50.0% | 0.5751 | 0.5000 | -0.0751 | 0.25930 | 0.71205 | 2 |
| ATS | FOUR-TIER-V1 | ELITE | 0.6000 | 2 | 2-0-0-0 | 100.0% | 0.8848 | 1.0000 | +0.1152 | 0.01405 | 0.12289 | 2 |
| ATS | SHARED-51.75 | WEAK | — | 108 | 58-50-0-0 | 53.7% | 0.5085 | 0.5370 | +0.0285 | 0.24948 | 0.69211 | 108 |
| ATS | SHARED-51.75 | MODERATE | 0.5175 | 113 | 57-56-0-0 | 50.4% | 0.5260 | 0.5044 | -0.0216 | 0.25122 | 0.69558 | 113 |
| ATS | SHARED-51.75 | STRONG | 0.5375 | 16 | 11-5-0-0 | 68.8% | 0.5429 | 0.6875 | +0.1446 | 0.23663 | 0.66635 | 16 |
| ATS | SHARED-51.75 | ELITE | 0.5575 | 1 | 0-1-0-0 | 0.0% | 0.5592 | 0.0000 | -0.5592 | 0.31272 | 0.81919 | 1 |
| ATS | SHARED-52/54/58 | WEAK | — | 65 | 33-32-0-0 | 50.8% | 0.5099 | 0.5077 | -0.0023 | 0.25004 | 0.69324 | 65 |
| ATS | SHARED-52/54/58 | MODERATE | 0.5200 | 45 | 18-27-0-0 | 40.0% | 0.5296 | 0.4000 | -0.1296 | 0.25715 | 0.70746 | 45 |
| ATS | SHARED-52/54/58 | STRONG | 0.5400 | 15 | 11-4-0-0 | 73.3% | 0.5486 | 0.7333 | +0.1847 | 0.22699 | 0.64693 | 15 |
| ATS | SHARED-52/54/58 | ELITE | 0.5800 | 0 | 0-0-0-0 | N/A | N/A | N/A | N/A | N/A | N/A | 0 |
| ATS | ENGINE-V2 | WEAK | — | 30 | 13-17-0-0 | 43.3% | 0.5060 | 0.4333 | -0.0727 | 0.25171 | 0.69658 | 30 |
| ATS | ENGINE-V2 | MODERATE | 0.5120 | 21 | 10-11-0-0 | 47.6% | 0.5193 | 0.4762 | -0.0431 | 0.25157 | 0.69629 | 21 |
| ATS | ENGINE-V2 | STRONG | 0.5318 | 4 | 0-4-0-0 | 0.0% | 0.5377 | 0.0000 | -0.5377 | 0.28908 | 0.77147 | 4 |
| ATS | ENGINE-V2 | ELITE | 0.5460 | 1 | 1-0-0-0 | 100.0% | 0.5488 | 1.0000 | +0.4512 | 0.20355 | 0.59995 | 1 |
| ATS | SLATE-QUARTILES-V3 | WEAK | — | 14 | 8-6-0-0 | 57.1% | 0.5040 | 0.5714 | +0.0674 | 0.24811 | 0.68937 | 14 |
| ATS | SLATE-QUARTILES-V3 | MODERATE | slate-specific | 14 | 10-4-0-0 | 71.4% | 0.5102 | 0.7143 | +0.2041 | 0.24551 | 0.68417 | 14 |
| ATS | SLATE-QUARTILES-V3 | STRONG | slate-specific | 14 | 7-7-0-0 | 50.0% | 0.5170 | 0.5000 | -0.0170 | 0.24958 | 0.69231 | 14 |
| ATS | SLATE-QUARTILES-V3 | ELITE | slate-specific | 13 | 8-5-0-0 | 61.5% | 0.5261 | 0.6154 | +0.0893 | 0.24520 | 0.68355 | 13 |
| TOTALS | FOUR-TIER-V1 | WEAK | — | 10 | 6-4-0-0 | 60.0% | 0.5116 | 0.6000 | +0.0884 | 0.25087 | 0.69488 | 10 |
| TOTALS | FOUR-TIER-V1 | MODERATE | 0.5300 | 3 | 0-3-0-0 | 0.0% | 0.5419 | 0.0000 | -0.5419 | 0.29367 | 0.78071 | 3 |
| TOTALS | FOUR-TIER-V1 | STRONG | 0.5600 | 1 | 0-1-0-0 | 0.0% | 0.5634 | 0.0000 | -0.5634 | 0.31743 | 0.82876 | 1 |
| TOTALS | FOUR-TIER-V1 | ELITE | 0.6000 | 5 | 4-1-0-0 | 80.0% | 0.7078 | 0.8000 | +0.0922 | 0.21670 | 0.64984 | 5 |
| TOTALS | SHARED-51.75 | WEAK | — | 87 | 50-37-0-0 | 57.5% | 0.5091 | 0.5747 | +0.0657 | 0.24878 | 0.69070 | 87 |
| TOTALS | SHARED-51.75 | MODERATE | 0.5175 | 77 | 38-39-0-0 | 49.4% | 0.5266 | 0.4935 | -0.0331 | 0.25186 | 0.69688 | 77 |
| TOTALS | SHARED-51.75 | STRONG | 0.5375 | 55 | 26-29-0-0 | 47.3% | 0.5457 | 0.4727 | -0.0730 | 0.25595 | 0.70510 | 55 |
| TOTALS | SHARED-51.75 | ELITE | 0.5575 | 19 | 14-5-0-0 | 73.7% | 0.5761 | 0.7368 | +0.1608 | 0.22128 | 0.63525 | 19 |
| TOTALS | SHARED-52/54/58 | WEAK | — | 60 | 32-28-0-0 | 53.3% | 0.5095 | 0.5333 | +0.0238 | 0.25008 | 0.69331 | 60 |
| TOTALS | SHARED-52/54/58 | MODERATE | 0.5200 | 40 | 15-25-0-0 | 37.5% | 0.5277 | 0.3750 | -0.1527 | 0.25910 | 0.71137 | 40 |
| TOTALS | SHARED-52/54/58 | STRONG | 0.5400 | 25 | 11-14-0-0 | 44.0% | 0.5521 | 0.4400 | -0.1121 | 0.25977 | 0.71280 | 25 |
| TOTALS | SHARED-52/54/58 | ELITE | 0.5800 | 0 | 0-0-0-0 | N/A | N/A | N/A | N/A | N/A | N/A | 0 |
| TOTALS | ENGINE-V2 | WEAK | — | 23 | 13-10-0-0 | 56.5% | 0.5028 | 0.5652 | +0.0624 | 0.24990 | 0.69295 | 23 |
| TOTALS | ENGINE-V2 | MODERATE | 0.5044 | 22 | 10-12-0-0 | 45.5% | 0.5071 | 0.4545 | -0.0526 | 0.25142 | 0.69599 | 22 |
| TOTALS | ENGINE-V2 | STRONG | 0.5111 | 11 | 8-3-0-0 | 72.7% | 0.5130 | 0.7273 | +0.2143 | 0.24356 | 0.68027 | 11 |
| TOTALS | ENGINE-V2 | ELITE | 0.5209 | 0 | 0-0-0-0 | N/A | N/A | N/A | N/A | N/A | N/A | 0 |
| TOTALS | SLATE-QUARTILES-V3 | WEAK | — | 14 | 5-9-0-0 | 35.7% | 0.5016 | 0.3571 | -0.1444 | 0.25093 | 0.69501 | 14 |
| TOTALS | SLATE-QUARTILES-V3 | MODERATE | slate-specific | 14 | 8-6-0-0 | 57.1% | 0.5042 | 0.5714 | +0.0673 | 0.24953 | 0.69220 | 14 |
| TOTALS | SLATE-QUARTILES-V3 | STRONG | slate-specific | 14 | 7-7-0-0 | 50.0% | 0.5082 | 0.5000 | -0.0082 | 0.24964 | 0.69242 | 14 |
| TOTALS | SLATE-QUARTILES-V3 | ELITE | slate-specific | 13 | 6-7-0-0 | 46.2% | 0.5122 | 0.4615 | -0.0507 | 0.25237 | 0.69789 | 13 |

Ratings classify the issued probability; current prospective evidence does not establish monotonic realized performance across rating bands. Spearman rank correlation of band rank with realized W/L, within each engine and definition: ATS FOUR-TIER-V1 -0.053001 (N=19); ATS SHARED-51.75 0.004591 (N=238); ATS SHARED-52/54/58 0.032005 (N=125); ATS ENGINE-V2 -0.041355 (N=56); ATS SLATE-QUARTILES-V3 -0.020523 (N=55); TOTALS FOUR-TIER-V1 0.021071 (N=19); TOTALS SHARED-51.75 -0.014188 (N=238); TOTALS SHARED-52/54/58 -0.109786 (N=125); TOTALS ENGINE-V2 0.061011 (N=56); TOTALS SLATE-QUARTILES-V3 0.056860 (N=55). Descriptive only.

## F. INTEGRITY SUMMARY

Counts are computed from the displayed rows. Line, price, probability and market-record checks compare each sealed position with the proposition its immutable grade settled.

| CHECK | TARGET SLATE | PHYSICAL CORPUS | EXPECTED | NOTE |
|---|---|---|---|---|
| ISSUED | 6 | 1018 | — | — |
| SETTLED | 6 | 986 | — | — |
| PENDING | 0 | 0 | ≥0 | official result not final; settles automatically |
| VOID | 0 | 0 | ≥0 | voided under the applicable FanDuel rule |
| PUSH | 0 | 0 | ≥0 | line landed exactly on the result |
| EXCLUDED POST KICKOFF | 0 | 32 | ≥0 | issued after kickoff; visible, never scored |
| DUPLICATES | 0 | 0 | 0 | — |
| MISSING GRADES | 0 | 0 | ≥0 | pending rows without a grade object (expected while a game is not final) |
| MISSING ISSUANCE EVIDENCE | 0 | 0 | 0 | — |
| LINE MISMATCHES | 0 | 0 | 0 | — |
| PRICE MISMATCHES | 0 | 0 | 0 | — |
| PROBABILITY MISMATCHES | 0 | 0 | 0 | — |
| MARKET RECORD MISMATCHES | 0 | 0 | 0 | — |
| T2 SNAPSHOT MISMATCHES | 0 | 0 | 0 | grade settled against a T-2 snapshot other than the one sealed on the card |
| MODEL HASH MISMATCHES | 0 | 0 | 0 | positions whose sealed engine hash differs from their own slate and engine |
| ENGINE EPOCHS | 2 | 34 | — | distinct sealed (engine, hash) pairs; earlier epochs are retained as issued (Appendix 4) |
| TARGET-SLATE ATS ENGINE = INSTALLED LOCKED ENGINE | DIFFERS | — | MATCH | installed locked engine is not the engine sealed on the target slate |
| TARGET-SLATE TOTALS ENGINE = INSTALLED LOCKED ENGINE | DIFFERS | — | MATCH | installed locked engine is not the engine sealed on the target slate |

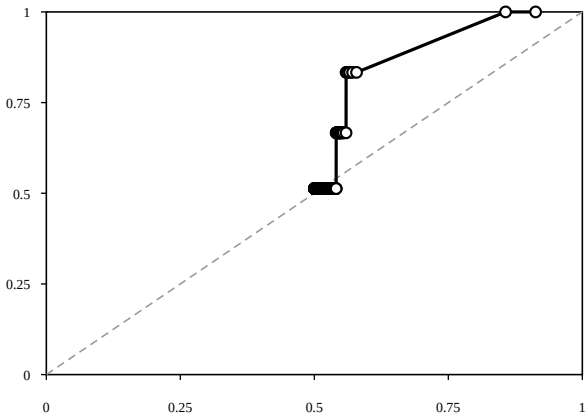
ENGINE_RERUN_COUNT=0 · CLOSING_LINE_SUBSTITUTION_COUNT=0 · MODEL_WRITE_COUNT=0 · ISSUANCE_WRITE_COUNT=0 · SETTLEMENT_WRITE_COUNT=0 · POSTGAME_FEATURE_BACKFILL_COUNT=0

KNOWN EVIDENCE LIMITATIONS=1 · FANDUEL_SETTLEMENT_RULE_SOURCE=UNRECORDED_JURISDICTION_DETAIL (stated under HOW TO READ).

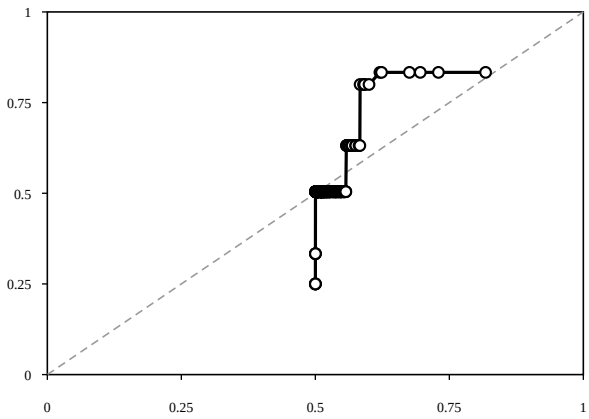
# APEX NCAA FOOTBALL - DAILY GRADER — TECHNICAL APPENDIX

## APPENDIX 1 — CALIBRATION AND DISCRIMINATION DIAGNOSTICS (RECORD ELIGIBLE W/L ROWS)

**ATS CORP / PAV RELIABILITY**



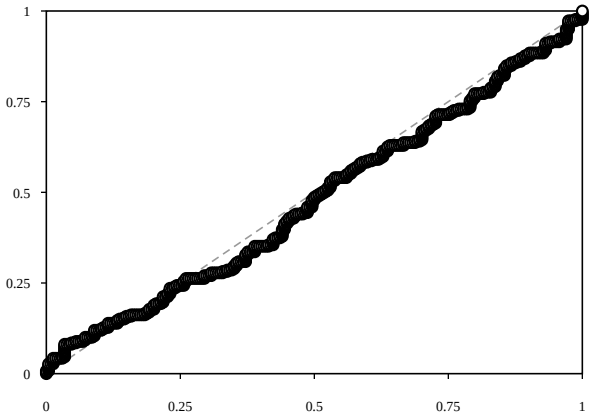

**TOTALS CORP / PAV RELIABILITY**



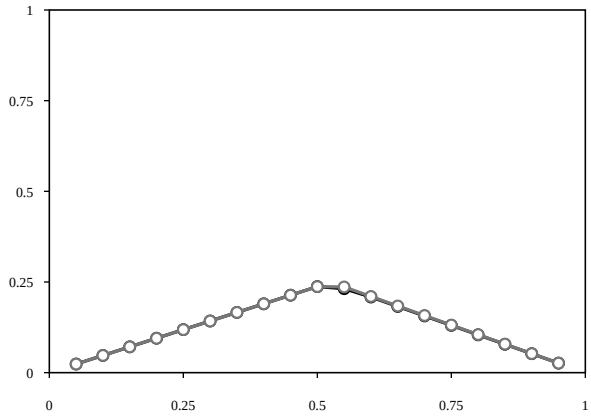

| ENGINE | N | BRIER | MISCALIBRATION | DISCRIMINATION | UNCERTAINTY | IDENTITY RESIDUAL |
|---|---|---|---|---|---|---|
| ATS | 493 | 0.2490059 | 0.0016648 | 0.0030642 | 0.2493571 | 0.0010483 |
| TOTALS | 493 | 0.2509374 | 0.0020144 | 0.0034524 | 0.2498262 | 0.0025492 |

CORP/PAV is a reliability diagnostic and proper-score decomposition on N=493 ATS and N=493 TOTALS record eligible W/L rows; descriptive, no significance claim.

**ATS ROC**




**ATS MURPHY**




ATS ROC AUC=0.479144 (N=493) · TOTALS AUC=0.467375 (N=493) · discrimination diagnostics only, not proper scores. Murphy: elementary-score / decision-threshold diagnostic, lower is better.

## APPENDIX 2 — SLICES: DIRECTION, LINE RANGE, TOTALS DISTRIBUTION

| ATS DEFINITION · BAND | FAVORITE N/W/L/P | UNDERDOG N/W/L/P | LOGLOSS | BRIER | SCORE N |
|---|---|---|---|---|---|
| FOUR-TIER-V1 · WEAK | 8/6/2/0 | 5/4/1/0 | 0.69199 | 0.24942 | 13 |
| FOUR-TIER-V1 · MODERATE | 2/1/1/0 | 0/0/0/0 | 0.71788 | 0.26227 | 2 |
| FOUR-TIER-V1 · STRONG | 0/0/0/0 | 2/1/1/0 | 0.71205 | 0.25930 | 2 |
| FOUR-TIER-V1 · ELITE | 0/0/0/0 | 2/2/0/0 | 0.12289 | 0.01405 | 2 |
| SHARED-51.75 · WEAK | 81/47/34/0 | 27/11/16/0 | 0.69211 | 0.24948 | 108 |
| SHARED-51.75 · MODERATE | 95/49/46/0 | 18/8/10/0 | 0.69558 | 0.25122 | 113 |
| SHARED-51.75 · STRONG | 11/8/3/0 | 5/3/2/0 | 0.66635 | 0.23663 | 16 |
| SHARED-51.75 · ELITE | 0/0/0/0 | 1/0/1/0 | 0.81919 | 0.31272 | 1 |
| SHARED-52/54/58 · WEAK | 41/22/19/0 | 24/11/13/0 | 0.69324 | 0.25004 | 65 |
| SHARED-52/54/58 · MODERATE | 28/10/18/0 | 17/8/9/0 | 0.70746 | 0.25715 | 45 |
| SHARED-52/54/58 · STRONG | 11/8/3/0 | 4/3/1/0 | 0.64693 | 0.22699 | 15 |
| SHARED-52/54/58 · ELITE | 0/0/0/0 | 0/0/0/0 | N/A | N/A | 0 |
| ENGINE-V2 · WEAK | 19/7/12/0 | 11/6/5/0 | 0.69658 | 0.25171 | 30 |
| ENGINE-V2 · MODERATE | 15/7/8/0 | 6/3/3/0 | 0.69629 | 0.25157 | 21 |
| ENGINE-V2 · STRONG | 4/0/4/0 | 0/0/0/0 | 0.77147 | 0.28908 | 4 |
| ENGINE-V2 · ELITE | 1/1/0/0 | 0/0/0/0 | 0.59995 | 0.20355 | 1 |
| SLATE-QUARTILES-V3 · WEAK | 7/4/3/0 | 7/4/3/0 | 0.68937 | 0.24811 | 14 |
| SLATE-QUARTILES-V3 · MODERATE | 3/2/1/0 | 11/8/3/0 | 0.68417 | 0.24551 | 14 |
| SLATE-QUARTILES-V3 · STRONG | 9/3/6/0 | 5/4/1/0 | 0.69231 | 0.24958 | 14 |
| SLATE-QUARTILES-V3 · ELITE | 8/5/3/0 | 5/3/2/0 | 0.68355 | 0.24520 | 13 |

| TOTALS DEFINITION · BAND | OVER N/W/L/P | UNDER N/W/L/P | PMF LOGSCORE | RPS | PMF N |
|---|---|---|---|---|---|
| FOUR-TIER-V1 · WEAK | 2/1/1/0 | 8/5/3/0 | 3.907464 | 0.029001 | 10 |
| FOUR-TIER-V1 · MODERATE | 0/0/0/0 | 3/0/3/0 | 4.510078 | 0.054096 | 3 |
| FOUR-TIER-V1 · STRONG | 0/0/0/0 | 1/0/1/0 | 3.951097 | 0.022986 | 1 |
| FOUR-TIER-V1 · ELITE | 0/0/0/0 | 5/4/1/0 | 3.913484 | 0.024155 | 5 |
| SHARED-51.75 · WEAK | 39/25/14/0 | 48/25/23/0 | 4.090637 | 0.033752 | 87 |
| SHARED-51.75 · MODERATE | 15/12/3/0 | 62/26/36/0 | 4.016740 | 0.032025 | 77 |
| SHARED-51.75 · STRONG | 6/4/2/0 | 49/22/27/0 | 4.149179 | 0.037101 | 55 |
| SHARED-51.75 · ELITE | 0/0/0/0 | 19/14/5/0 | 4.052026 | 0.032149 | 19 |
| SHARED-52/54/58 · WEAK | 39/21/18/0 | 21/11/10/0 | 4.088244 | 0.034117 | 60 |
| SHARED-52/54/58 · MODERATE | 14/5/9/0 | 26/10/16/0 | 4.107484 | 0.036594 | 40 |
| SHARED-52/54/58 · STRONG | 10/6/4/0 | 15/5/10/0 | 4.016503 | 0.031362 | 25 |
| SHARED-52/54/58 · ELITE | 0/0/0/0 | 0/0/0/0 | N/A | N/A | 0 |
| ENGINE-V2 · WEAK | 16/7/9/0 | 7/6/1/0 | 4.074085 | 0.033889 | 20 |
| ENGINE-V2 · MODERATE | 19/8/11/0 | 3/2/1/0 | 4.173835 | 0.033637 | 22 |
| ENGINE-V2 · STRONG | 11/8/3/0 | 0/0/0/0 | 3.860586 | 0.028300 | 11 |
| ENGINE-V2 · ELITE | 0/0/0/0 | 0/0/0/0 | N/A | N/A | 0 |

| TOTALS DEFINITION · BAND | OVER N/W/L/P | UNDER N/W/L/P | PMF LOGSCORE | RPS | PMF N |
|---|---|---|---|---|---|
| SLATE-QUARTILES-V3 · WEAK | 8/2/6/0 | 6/3/3/0 | 4.203457 | 0.043574 | 14 |
| SLATE-QUARTILES-V3 · MODERATE | 8/6/2/0 | 6/2/4/0 | 4.407191 | 0.044611 | 14 |
| SLATE-QUARTILES-V3 · STRONG | 11/6/5/0 | 3/1/2/0 | 4.113432 | 0.035804 | 14 |
| SLATE-QUARTILES-V3 · ELITE | 13/6/7/0 | 0/0/0/0 | 3.816391 | 0.028162 | 13 |

TARGET-SLATE RETAINED TOTALS PMFs=3 (maximum normalization error 0.000e+00) · PROSPECTIVE PMF-SCORED TOTALS POSITIONS=490. No PMF was inferred from moments or binary probabilities.

| ENGINE | FAMILY | SLICE | N | W-L-P | WIN % | MEAN P | EMPIRICAL | LOGLOSS | BRIER | SCORE N |
|---|---|---|---|---|---|---|---|---|---|---|
| ATS | DIRECTION | FAVORITE | 343 | 180-163-0 | 52.5% | 0.5195 | 0.5248 | 0.69286 | 0.24986 | 343 |
| ATS | DIRECTION | UNDERDOG | 150 | 79-71-0 | 52.7% | 0.5219 | 0.5267 | 0.68595 | 0.24706 | 150 |
| ATS | SPREAD_SIZE | 0-3 | 64 | 25-39-0 | 39.1% | 0.5209 | 0.3906 | 0.70080 | 0.25383 | 64 |
| ATS | SPREAD_SIZE | 14+ | 230 | 136-94-0 | 59.1% | 0.5218 | 0.5913 | 0.68386 | 0.24579 | 230 |
| ATS | SPREAD_SIZE | 3.5-7 | 100 | 47-53-0 | 47.0% | 0.5186 | 0.4700 | 0.69821 | 0.25253 | 100 |
| ATS | SPREAD_SIZE | 7.5-14 | 99 | 51-48-0 | 51.5% | 0.5179 | 0.5152 | 0.69277 | 0.24981 | 99 |
| TOTALS | DIRECTION | OVER | 211 | 117-94-0 | 55.5% | 0.5136 | 0.5545 | 0.69028 | 0.24857 | 211 |
| TOTALS | DIRECTION | UNDER | 282 | 136-146-0 | 48.2% | 0.5318 | 0.4823 | 0.69898 | 0.25271 | 282 |
| TOTALS | TOTAL_SIZE | 45-52.5 | 230 | 130-100-0 | 56.5% | 0.5127 | 0.5652 | 0.69117 | 0.24902 | 230 |
| TOTALS | TOTAL_SIZE | 53-59.5 | 184 | 84-100-0 | 45.7% | 0.5297 | 0.4565 | 0.69425 | 0.25068 | 184 |
| TOTALS | TOTAL_SIZE | 60+ | 41 | 18-23-0 | 43.9% | 0.5593 | 0.4390 | 0.72724 | 0.26497 | 41 |
| TOTALS | TOTAL_SIZE | <45 | 38 | 21-17-0 | 55.3% | 0.5267 | 0.5526 | 0.69040 | 0.24863 | 38 |

# APPENDIX 3 — WEEKLY AND ROLLING EVIDENCE

| SEASON-WEEK | ENGINE | N | W | L | P | V | PEND | LOGLOSS | BRIER | SCORE N |
|---|---|---|---|---|---|---|---|---|---|---|
| 2026-W1 | ATS | 98 | 58 | 40 | 0 | 0 | 0 | 0.681548 | 0.245199 | 98 |
| 2026-W1 | TOTALS | 98 | 50 | 48 | 0 | 0 | 0 | 0.698386 | 0.252019 | 98 |
| 2026-W2 | ATS | 86 | 47 | 39 | 0 | 0 | 0 | 0.691017 | 0.248935 | 86 |
| 2026-W2 | TOTALS | 86 | 46 | 40 | 0 | 0 | 0 | 0.690113 | 0.248491 | 86 |
| 2026-W3 | ATS | 73 | 35 | 38 | 0 | 0 | 0 | 0.694683 | 0.250766 | 73 |
| 2026-W3 | TOTALS | 73 | 42 | 31 | 0 | 0 | 0 | 0.687370 | 0.247131 | 73 |
| 2026-W4 | ATS | 54 | 24 | 30 | 0 | 0 | 0 | 0.697022 | 0.251941 | 54 |
| 2026-W4 | TOTALS | 54 | 28 | 26 | 0 | 0 | 0 | 0.699773 | 0.253289 | 54 |
| 2026-W5 | ATS | 59 | 32 | 27 | 0 | 0 | 0 | 0.689044 | 0.247959 | 59 |
| 2026-W5 | TOTALS | 59 | 24 | 35 | 0 | 0 | 0 | 0.705401 | 0.256117 | 59 |
| 2026-W6 | ATS | 58 | 25 | 33 | 0 | 0 | 0 | 0.698314 | 0.252583 | 58 |
| 2026-W6 | TOTALS | 58 | 30 | 28 | 0 | 0 | 0 | 0.695412 | 0.251131 | 58 |
| 2026-W7 | ATS | 62 | 35 | 27 | 0 | 0 | 0 | 0.690855 | 0.248854 | 62 |
| 2026-W7 | TOTALS | 62 | 31 | 31 | 0 | 0 | 0 | 0.693489 | 0.250171 | 62 |
| 2026-W8 | ATS | 3 | 3 | 0 | 0 | 0 | 0 | 0.661762 | 0.234314 | 3 |
| 2026-W8 | TOTALS | 3 | 2 | 1 | 0 | 0 | 0 | 0.685623 | 0.246238 | 3 |

| WINDOW | STATUS | SLATES | N | W | L | P | V | PENDING |
|---|---|---|---|---|---|---|---|---|
| ROLLING_LATEST_5 | AVAILABLE | 5 | 126 | 67 | 59 | 0 | 0 | 0 |
| ROLLING_LATEST_10 | AVAILABLE | 10 | 244 | 125 | 119 | 0 | 0 | 0 |
| ROLLING_LATEST_20 | AVAILABLE | 20 | 618 | 311 | 307 | 0 | 0 | 0 |

# APPENDIX 4 — CORPUS COVERAGE, EPOCHS AND HASHES

| SLATE | PHYSICAL | EXCLUDED | DIRECTIONAL | EXACT P | EXACT PMF | FEATURE VECTOR | PENDING | VOID |
|---|---|---|---|---|---|---|---|---|
| 2026-08-29 | 16 | 0 | 16 | 16 | 16 | 0 | 0 | 0 |
| 2026-09-03 | 22 | 0 | 22 | 22 | 22 | 22 | 0 | 0 |
| 2026-09-04 | 14 | 0 | 14 | 14 | 14 | 14 | 0 | 0 |
| 2026-09-05 | 136 | 0 | 136 | 136 | 136 | 136 | 0 | 0 |
| 2026-09-06 | 6 | 0 | 6 | 6 | 6 | 6 | 0 | 0 |
| 2026-09-07 | 2 | 0 | 2 | 2 | 2 | 2 | 0 | 0 |
| 2026-09-10 | 2 | 0 | 2 | 2 | 2 | 2 | 0 | 0 |
| 2026-09-11 | 10 | 0 | 10 | 10 | 10 | 10 | 0 | 0 |
| 2026-09-12 | 160 | 0 | 160 | 160 | 160 | 160 | 0 | 0 |
| 2026-09-17 | 2 | 0 | 2 | 2 | 2 | 2 | 0 | 0 |
| 2026-09-18 | 6 | 0 | 6 | 6 | 6 | 6 | 0 | 0 |
| 2026-09-19 | 138 | 0 | 138 | 138 | 138 | 138 | 0 | 0 |
| 2026-09-24 | 2 | 0 | 2 | 2 | 2 | 2 | 0 | 0 |
| 2026-09-25 | 10 | 2 | 10 | 10 | 10 | 10 | 0 | 0 |
| 2026-09-26 | 128 | 30 | 128 | 128 | 128 | 128 | 0 | 0 |
| 2026-10-01 | 4 | 0 | 4 | 4 | 4 | 4 | 0 | 0 |
| 2026-10-02 | 6 | 0 | 6 | 6 | 6 | 6 | 0 | 0 |
| 2026-10-03 | 108 | 0 | 108 | 108 | 108 | 108 | 0 | 0 |
| 2026-10-06 | 2 | 0 | 2 | 2 | 2 | 2 | 0 | 0 |
| 2026-10-07 | 4 | 0 | 4 | 4 | 4 | 4 | 0 | 0 |
| 2026-10-08 | 8 | 0 | 8 | 8 | 8 | 8 | 0 | 0 |
| 2026-10-09 | 10 | 0 | 10 | 10 | 10 | 10 | 0 | 0 |
| 2026-10-10 | 92 | 0 | 92 | 92 | 92 | 92 | 0 | 0 |
| 2026-10-13 | 4 | 0 | 4 | 4 | 4 | 4 | 0 | 0 |
| 2026-10-14 | 4 | 0 | 4 | 4 | 4 | 4 | 0 | 0 |
| 2026-10-15 | 6 | 0 | 6 | 6 | 0 | 0 | 0 | 0 |
| 2026-10-16 | 6 | 0 | 6 | 6 | 6 | 6 | 0 | 0 |
| 2026-10-17 | 104 | 0 | 104 | 104 | 104 | 104 | 0 | 0 |
| 2026-10-20 | 6 | 0 | 6 | 6 | 6 | 6 | 0 | 0 |

Only physical immutable production evidence is listed; no slate is synthesized. EXCLUDED counts positions issued post-kickoff on that slate.

| SLATE | ATS MODEL | ATS SHA256 | TOTALS MODEL | TOTALS SHA256 | ISSUANCE SHA256 | POS |
|---|---|---|---|---|---|---|
| 2026-08-29 | GLOBAL_EMPIRICAL_V1 | 2ac5790dca0f91da5431f44fb4262ef0398f2adb1897cb9ded311fb4470c8798 | RIDGE_GAUSSIAN_A100_V1 | 7524bfe56bc65257e5257a5e751540ed29bb6ca027b5193af87db35aaa79da9b | 7ca849afef6c528d1ead8d7437d69de4c2d0495fa17ae6a7cc149b752b7fb5cf | 16 |
| 2026-09-03 | D2 | fc79ff17b8de2a00019e10f4dc8c2f5bb16acf14648fcb65a31decc3d9368494 | TOTALS-V5-RECONCILED-POOL | d22ea083819b70c22f409a032ddab1d03322c10790541c8d458266868e7835e8 | 206e29bdc0684f4fb28c38e4d31fef213044b0e2b5921244ee6c95f59bf577f5 | 22 |
| 2026-09-04 | D2 | 61bde29368eeadbad93677743726c0ac7e6ae58e1607ca0efeae6be28791af8a | TOTALS-V5-RECONCILED-POOL | 1949c669883a6de379e866a7955ba414526a74edcecd9980f56cb67774ba15a4 | 60d17f19f198dacf58b27b9effa827a2a218da8ee50d8167358bc6e5a96ff77b | 14 |
| 2026-09-05 | D2 | fbaf65a5d6c42a1dd3eb896fa6448101ac18ae87a2c725208597639a13f8c28a | TOTALS-V5-RECONCILED-POOL | a44b3556bb82badf472033ad582e3872b74eef8c16b880dd83bc564cb866393e | d4e843d2045430ab571451618707fc263e098730c11387f78f18216d8a9db317 | 136 |
| 2026-09-06 | D2 | 6b1b00207e63d1bfc02fb5b0e0a8e10d43d65b75e507b5d779281598306d03c4 | TOTALS-V5-RECONCILED-POOL | 3c8333e071f531230eb6df70c771efca66e2502be153f12860590952c88e104f | 86db9039d44d631ce1b1c4d2634b245ee4de57f6e7c16320bb0b4c1b0eb57638 | 6 |
| 2026-09-07 | D2 | 9d2e7e3fd1d851c4acfd6cae6c5936a6e2640f2ec47c1b9a6fd887ddda2505e7 | TOTALS-V5-RECONCILED-POOL | 4867d2e67a77260058b42c84ce61f53000ecebc861a72c0d226cc52261dd75c3 | 21ae7f8b7c21dc9dd8a2648037b759e41932cd41a56ace606a6c750071a6ba2f | 2 |
| 2026-09-10 | D2 | 9d2e7e3fd1d851c4acfd6cae6c5936a6e2640f2ec47c1b9a6fd887ddda2505e7 | TOTALS-V5-RECONCILED-POOL | 4867d2e67a77260058b42c84ce61f53000ecebc861a72c0d226cc52261dd75c3 | 092c007808e9eaf0621ad447487a4eb2edcc6b357dded58966dd80a699905465 | 2 |
| 2026-09-11 | D2 | 9d2e7e3fd1d851c4acfd6cae6c5936a6e2640f2ec47c1b9a6fd887ddda2505e7 | TOTALS-V5-RECONCILED-POOL | 4867d2e67a77260058b42c84ce61f53000ecebc861a72c0d226cc52261dd75c3 | 31dbf89d125019d978fc51ede5e2901cd580547ed9a92cb061dae348a880ebd9 | 10 |
| 2026-09-12 | D2 | 9d2e7e3fd1d851c4acfd6cae6c5936a6e2640f2ec47c1b9a6fd887ddda2505e7 | TOTALS-V5-RECONCILED-POOL | 4867d2e67a77260058b42c84ce61f53000ecebc861a72c0d226cc52261dd75c3 | ef2c885600d42f8ab9638655b55f6e660784633a619de2f4a2689f704cf1833a | 160 |
| 2026-09-17 | D2 | 9d2e7e3fd1d851c4acfd6cae6c5936a6e2640f2ec47c1b9a6fd887ddda2505e7 | TOTALS-V5-RECONCILED-POOL | 4867d2e67a77260058b42c84ce61f53000ecebc861a72c0d226cc52261dd75c3 | cd22addc00b553ae62058953be1695202952efdf5db43f4a47a6bd1993fbf491 | 2 |
| 2026-09-18 | D2 | 9d2e7e3fd1d851c4acfd6cae6c5936a6e2640f2ec47c1b9a6fd887ddda2505e7 | TOTALS-V5-RECONCILED-POOL | 4867d2e67a77260058b42c84ce61f53000ecebc861a72c0d226cc52261dd75c3 | a3ca7166267b456484096417ec3d0d10f67ec1fac36800fb4f3eee6280b70981 | 6 |
| 2026-09-19 | D2 | 9d2e7e3fd1d851c4acfd6cae6c5936a6e2640f2ec47c1b9a6fd887ddda2505e7 | TOTALS-V5-RECONCILED-POOL | 4867d2e67a77260058b42c84ce61f53000ecebc861a72c0d226cc52261dd75c3 | 0b08644781 2f9a3953a6dd7df666703859be26ec5e4c13f3d54469bc31c53e69 | 138 |
| 2026-09-24 | D2 | e46880f3483a7badb7329fee2c709eb1b63b1a250aea9dfaa2cd2fdc63606469 | TOTALS-V5-RECONCILED-POOL | 4a9b487849a1899e8ad11ff10fbce1e8c16243e6d8accdb95ff4de5d7cf26725 | aa1effee193bc769a1faa9561f968673c5f34c0767d22184b3b6cafdceec0f9c | 2 |
| 2026-09-25 | D2 | e46880f3483a7badb7329fee2c709eb1b63b1a250aea9dfaa2cd2fdc63606469 | TOTALS-V5-RECONCILED-POOL | 4a9b487849a1899e8ad11ff10fbce1e8c16243e6d8accdb95ff4de5d7cf26725 | f54aa7358ceb2c8c587ae954f77dd2316bf608a094e612857102cc0fa6c8bb50 | 10 |
| 2026-09-26 | D2 | 69d8a6c392ded2d2fb728b95ab69255b505709bce7cbfd9387314cefe666359c | TOTALS-V5-RECONCILED-POOL | 43a2ac80cd611e14793cdbd76d3c663457ea4f78cdfe9d4aed4ce0be30292f4b | bdcffed33a92cbe8a811365359aa59e800ffb5764b4d85ca4a734bb40c46dc9c8 | 128 |
| 2026-10-01 | D2 | 6553b110ab1f5610bb3e661030b51219729111e8bb17b0ec35b306307 50f9969 | TOTALS-V5-RECONCILED-POOL | a96560915ec8fe89648a872c06b463bf1e8ff911181210ca27661083e83e53f3 | dbeb13faabe6375ba5f07cf107e1f8d498c2da288a0fab60cf9f09c029573f0c | 4 |
| 2026-10-02 | D2 | 6553b110ab1f5610bb3e661030b51219729111e8bb17b0ec35b306307 50f9969 | TOTALS-V5-RECONCILED-POOL | a96560915ec8fe89648a872c06b463bf1e8ff911181210ca27661083e83e53f3 | 67b8ec6ef73f4a059133509d82312dd2d03be72013f901cedff7a266e5b9b5d2 | 6 |
| 2026-10-03 | D2 | 6553b110ab1f5610bb3e661030b51219729111e8bb17b0ec35b306307 50f9969 | TOTALS-V5-RECONCILED-POOL | a96560915ec8fe89648a872c06b463bf1e8ff911181210ca27661083e83e53f3 | 96ca5988359bc3786e4d2f39ec67da17fcf15cd3a11acca9da27f19942248637 | 108 |
| 2026-10-06 | D2 | 6553b110ab1f5610bb3e661030b51219729111e8bb17b0ec35b306307 50f9969 | TOTALS-V5-RECONCILED-POOL | a96560915ec8fe89648a872c06b463bf1e8ff911181210ca27661083e83e53f3 | a6f873de88fad0362a04c348c70cc6db1212c361f0db7e4ebb7f3460964bdaf7 | 2 |
| 2026-10-07 | D2 | 83c82036b5009db103d32cfe38078c5f906d60f076155bd7493f6b722c151c3d | TOTALS-V5-RECONCILED-POOL | 699cbaa1ab0a58c1ee0c215beaf1b59d093ff934c15733141b76ac725f890a7c | 6e6c1a1868dde2fdaebcc7edd866c3eb2893f0e79eabfbd1f3cca6210e18c0a3 | 4 |
| 2026-10-08 | D2 | 0e60ebc6cde767d75ce8811b2b3a660229145af0c76868a62f78b685ea654a33 | TOTALS-V5-RECONCILED-POOL | 9f586104744d101077a66b29a2bc3d8cec8ac28aa9bd5d5e003fb577b5bd71d2 | 5d4a2697f56d9d6387da5120ad482db3cdea2fa0e201c0d8dab08d491973c4b6 | 8 |
| 2026-10-09 | D2 | 39e779ec7f112d488a9379cac49ae70553895dfa2027df849a91ef512fcde8aa | TOTALS-V5-RECONCILED-POOL | 6a90e31cb2e12b0af43b74ce3a1f7cc52cf615993 94cbb3f90d89a753f50de81 | bec1312906c155872704c3df9cf35c1ac0940cd28254caf7177a0b533873885b | 10 |
| 2026-10-10 | D2 | 3844cb449793385ad6c30e02be76e8247ab5eb8f421fd8ac4631c4bdabc1d45c | TOTALS-V5-RECONCILED-POOL | 8e3f0746c12e8459e8a70ea4e41a36dc2fbccc085d7f7c93f38ce14eda497d26 | 7f0a3507742b30cf9f187b3f1a482158124a8bba747eaf80d8cc4e86c2745979 | 92 |
| 2026-10-13 | D2 | 3844cb449793385ad6c30e02be76e8247ab5eb8f421fd8ac4631c4bdabc1d45c | TOTALS-V5-RECONCILED-POOL | 8e3f0746c12e8459e8a70ea4e41a36dc2fbccc085d7f7c93f38ce14eda497d26 | e0ff93a241d1e5a9926e60356a0baecc8983d89728cccc4e2db5c8f7144555e2 | 4 |
| 2026-10-14 | D2 | 3844cb449793385ad6c30e02be76e8247ab5eb8f421fd8ac4631c4bdabc1d45c | TOTALS-V5-RECONCILED-POOL | 8e3f0746c12e8459e8a70ea4e41a36dc2fbccc085d7f7c93f38ce14eda497d26 | 1152153c9bdbbb876eff7e6fe0748570a39b49f50a17b63a45b286befe26ba89 | 4 |
| 2026-10-15 | D2 | — | TOTALS-V5-RECONCILED-POOL | — | 4ca57a804c77b104cf449e21460e68aea8faadead00b6bfa7b7fb5428ffdd11d | 6 |
| 2026-10-16 | D2 | c3c63d810d7b3d4b909db08afcb0a32cebdabb523f9488d5b61fb8f6854d9941 | TOTALS-V5-RECONCILED-POOL | 09bb7f9a3e2c3df5fae7d8d4893038f743a731eaefd5e3db6a595c391541e72f | d36c170767b2eaf6a5171d1ffd6c524d145b02d1ca7809473bfb58cea495c4ae | 6 |
| 2026-10-17 | D2 | 3bcb42aa283aff56f3a7b81743bffcbeeb16201ef4fee6577b3528cdb0761698 | TOTALS-V5-RECONCILED-POOL | ad4a7acc8fc64de5411af3cc3ff8395ad2d978aaee857ca5599323c276e8d029 | 4e324d5c8eb5773e15ddf6ae0a38193a2ed42471a588b49713d2e0541d403d07 | 104 |
| 2026-10-20 | D2 | 09c33b70af53b0a53e2127b16b00d23d37b8de692edc76f1ab83b1a25109d2e0 | TOTALS-V5-RECONCILED-POOL | e66c7f0d35a0827a93458b84e083d3d0fa4e54724af82470a674acb961c2243f | 0d15a3c6f93c346e5654127233681033508f8c3a2362d4e2576a3a16b873348c | 6 |

Each slate reports the engines physically sealed on that slate; earlier epochs are not relabelled with later champion identities.

# APPENDIX 5 — ISSUE-TIME EVIDENCE COVERAGE, CADENCE, DATA / CONTRACT RECEIPT AND SEQUENTIAL EVIDENCE

| OBJECT (TARGET SLATE) | CAPTURED | PHYSICAL COUNT | LIMITATION |
|---|---|---|---|
| ISSUED ATS PROBABILITY | YES | 3 | NONE |
| TOTALS FULL PMF | YES | 3 | RETAINED IN SEALED CARD |
| MARKET LINE / ISSUED PRICE | YES | 6 | NONE |
| FANDUEL QUOTE AT ISSUANCE (record sha, timestamps) | YES | 6 | NONE |
| MODEL ID / HASH · AS-ISSUED RATING | YES | 6 | NONE |
| D2 EXACT 348 VECTOR · V5 EXACT 353 VECTOR | YES | 3 | NONE |
| T3 / T2 SNAPSHOT HASHES | YES | 3 | NONE |
| ISSUE-TIME FEATURE SIDECAR | YES | 3 | NONE |
| SETTLEMENT | YES | 6 | IMMUTABLE GRADE AUTHORITY; POST-KICKOFF EXCLUDED |

D2_INPUTS_FILLED=792/810 · V5_INPUTS_FILLED=669/669 (filled/registered inputs the engines read, summed over this slate's champion cards). Champion cards retain exact vectors, missingness, source timestamps, availability state, full PMFs and FanDuel comparators; FCS fields the provider does not publish are retained as explicit missingness.

| CADENCE | PERMANENT NCAA REVIEW ROLE |
|---|---|
| DAILY | Immutable issuance-to-grade reconciliation; report products; delivery ledger |
| WEEKLY | Records, proper scores, direction and line-range slices with N |
| MONTHLY | Calibration, discrimination, band order, hash epochs and coverage |
| QUARTERLY | Scientific defensibility and prospective evidence review; no automatic retraining |

| OBJECT | PATH / ID · ROWS · SCHEMA | SHA256 |
|---|---|---|
| SEALED ISSUANCE | /var/opt/apex_ncaaf/production/issuance/2026-10-20/slate.json | 0d15a3c6f93c346e5654127233681033508f8c3a2362d4e2576a3a16b873348c |
| GRADE COMPLETION | /var/opt/apex_ncaaf/production/grader_completion_state/2026-10-20.json | 0b8e68f93b6f98d82de513d7d3b1b70dc95766173f8c9bac66b075af663511ad |
| ATS MODEL AT ISSUE | D2 | 09c33b70af53b0a53e2127b16b00d23d37b8de692edc76f1ab83b1a25109d2e0 |
| TOTALS MODEL AT ISSUE | TOTALS-V5-RECONCILED-POOL | e66c7f0d35a0827a93458b84e083d3d0fa4e54724af82470a674acb961c2243f |
| POSITION EVIDENCE | /var/opt/apex_ncaaf/production/results/2026-10-20/evidence/ncaaf_position_evidence.csv | 001b5d2ec881f8147f3d4817af13cdaf177632c569e42849591a36e45672e9f8 |
| COMPLETE SCIENTIFIC SIDECAR | /var/opt/apex_ncaaf/production/results/2026-10-20/evidence/ncaaf_complete_scientific_evidence.json · 1012 · apex.ncaaf.complete_position_scientific_evidence.2026-08-31.v1 | 97bb63aebf9c3f9111381df0a8c2397ac5f0b3d6dd9ca8f64a3fdb2ebea602ee |
| SEQUENTIAL FANDUEL SIDECAR | /var/opt/apex_ncaaf/production/results/2026-10-20/evidence/ncaaf_prospective_fanduel_sequential_evidence.json · 1900 · apex.ncaaf.prospective_fanduel_anytime_evidence.2026-09-04.v2 | 4010803a492e6139776722dbd544a8a0915529fdda39fa505726cc908b46dbf3 |
| COVERAGE LEDGER | /var/opt/apex_ncaaf/production/results/2026-10-20/evidence/ncaaf_coverage_ledger.csv | 2f7bc6d2d41710111d46c7063e543499f7745a02fb3bdd9e94fd2a18d49676a0 |


Brier (1950), DOI 10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2

Murphy (1969), DOI 10.1175/1520-0450(1969)008<0988:OTPS>2.0.CO;2

Gneiting & Raftery (2007), DOI 10.1198/016214506000001437

Dimitriadis, Gneiting & Jordan (2021), DOI 10.1073/PNAS.2016191118

Dimitriadis, Gneiting, Jordan & Vogel, DOI 10.1016/j.ijforecast.2023.09.007

Fawcett (2006), DOI 10.1016/j.patrec.2005.10.010


## PROSPECTIVE SEQUENTIAL FANDUEL EVIDENCE — CONFIDENCE SEQUENCE / E-PROCESS (RETAINED; INTERVALS NOT YET PRACTICALLY INFORMATIVE)

| ENGINE | METRIC | N | MEAN PAIRED ADVANTAGE (FD loss − APEX loss) | 95% CS LOW | 95% CS HIGH | ANYTIME P |
|---|---|---|---|---|---|---|
| D2 | BRIER | 475 | +0.000632 | -0.259065 | +0.260329 | 1.0000 |
| D2 | LOGLOSS | 475 | +0.001267 | -8.968353 | +8.970887 | 1.0000 |
| TOTALS-V5-RECONCILED-POOL | BRIER | 475 | -0.002082 | -0.261779 | +0.257615 | 1.0000 |
| TOTALS-V5-RECONCILED-POOL | LOGLOSS | 475 | -0.004164 | -8.973784 | +8.965456 | 1.0000 |

Never-reset bounded confidence sequence and mixture e-process over every grade the canonical prospective court retains: 950 sealed predictions, 950 graded, 1900 sequence rows, activation epoch 2026-09-01T03:30:00Z. The court admits only positions sealed on or after its activation epoch, so its N per engine is smaller than the paired N in section C, which pairs every record-eligible grade that carries a FanDuel comparator. Framework: BOUNDED_HOEFFDING_STITCHED_CONFIDENCE_SEQUENCE_AND_PREDICTABLE_MIXTURE_E_PROCESS.

BLACK_AND_WHITE_ONLY=YES · MODEL_WRITE_COUNT=0 · ISSUANCE_WRITE_COUNT=0 · SETTLEMENT_WRITE_COUNT=0 · ENGINE_RERUN_COUNT=0 · CLOSING_LINE_SUBSTITUTION_COUNT=0